Реализация чистой архитектуры. Ч.2

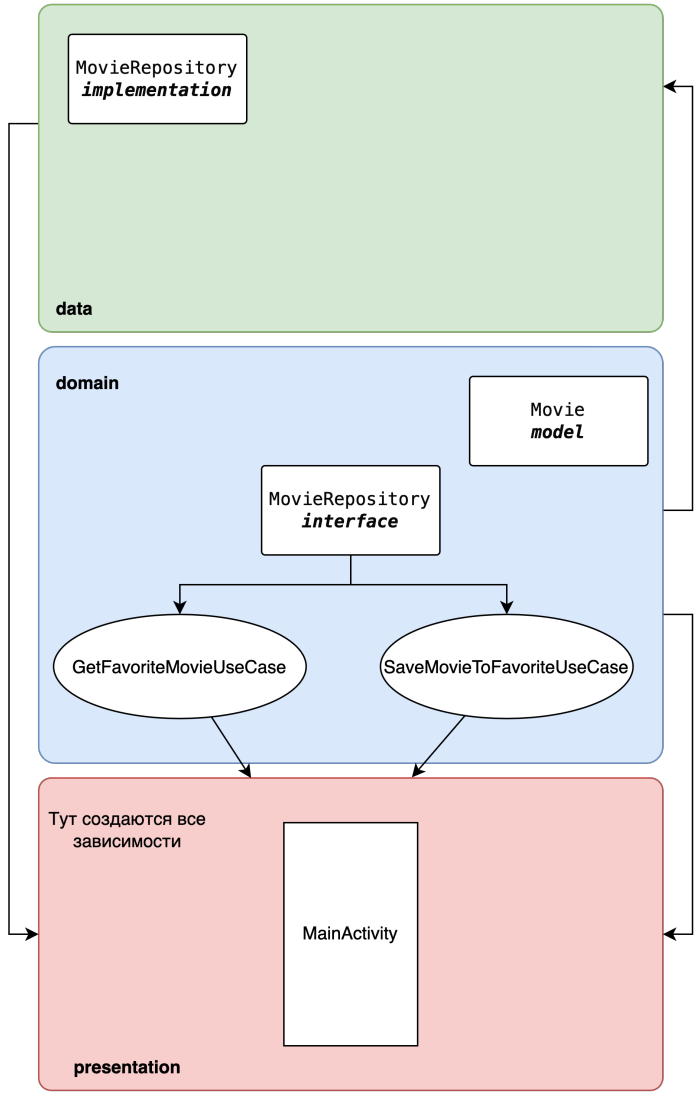
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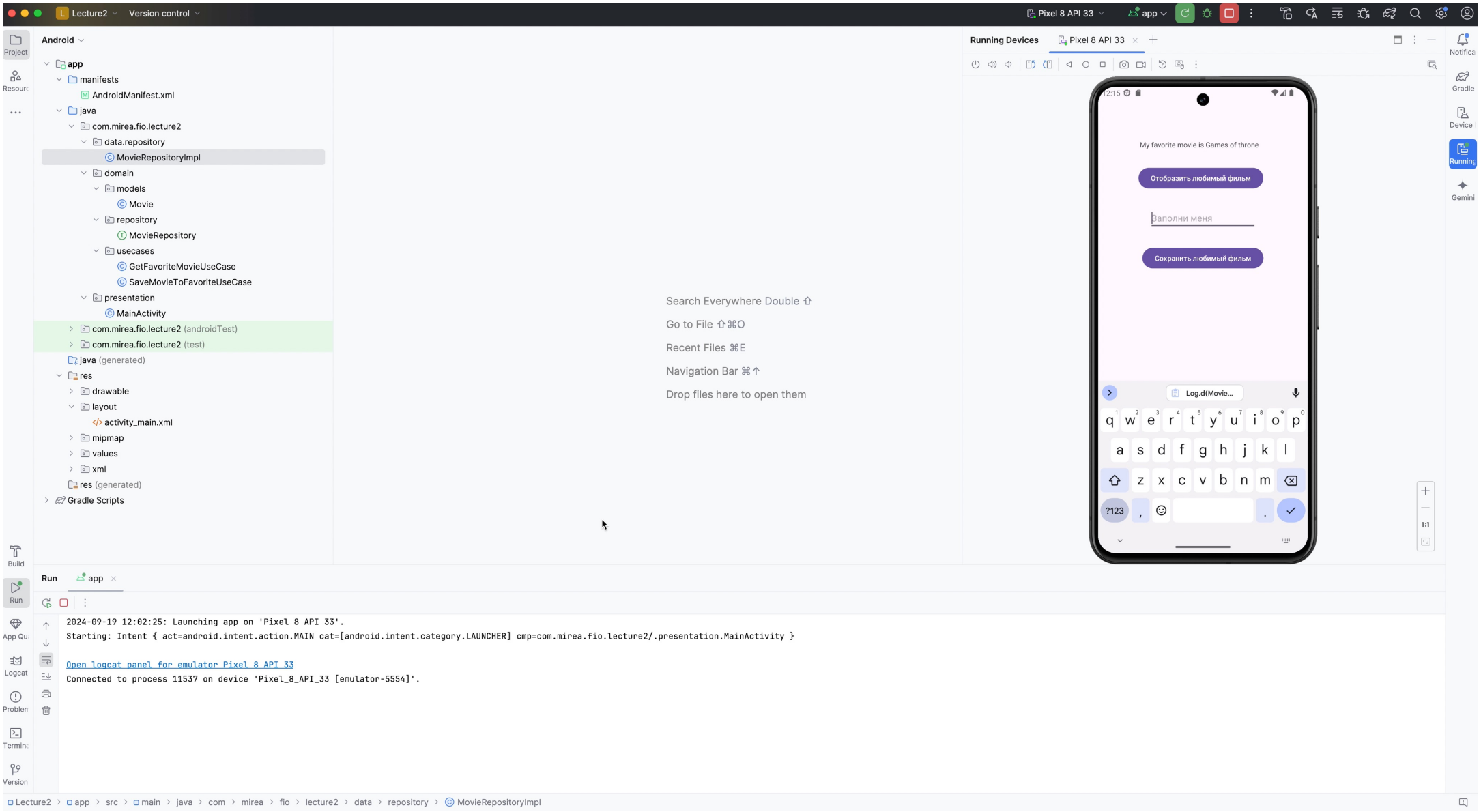
Чистая архитектура

Реализация

- app
  - manifests
  - java
    - com.mirea.fio.lecture1
      - data
        - model
        - repository
          - MovieRepositoryImpl
      - domain
        - models
          - Movie
        - repository
          - MovieRepository
        - usecases
          - GetFavoriteFilmUseCase
          - GetMoviesInfoByPageUseCase
          - LoginByEmailUseCase
          - RegisterByEmailUseCase
          - SaveMovieToFavoriteUseCase
      - presentation
        - MainActivity



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Чистая архитектура

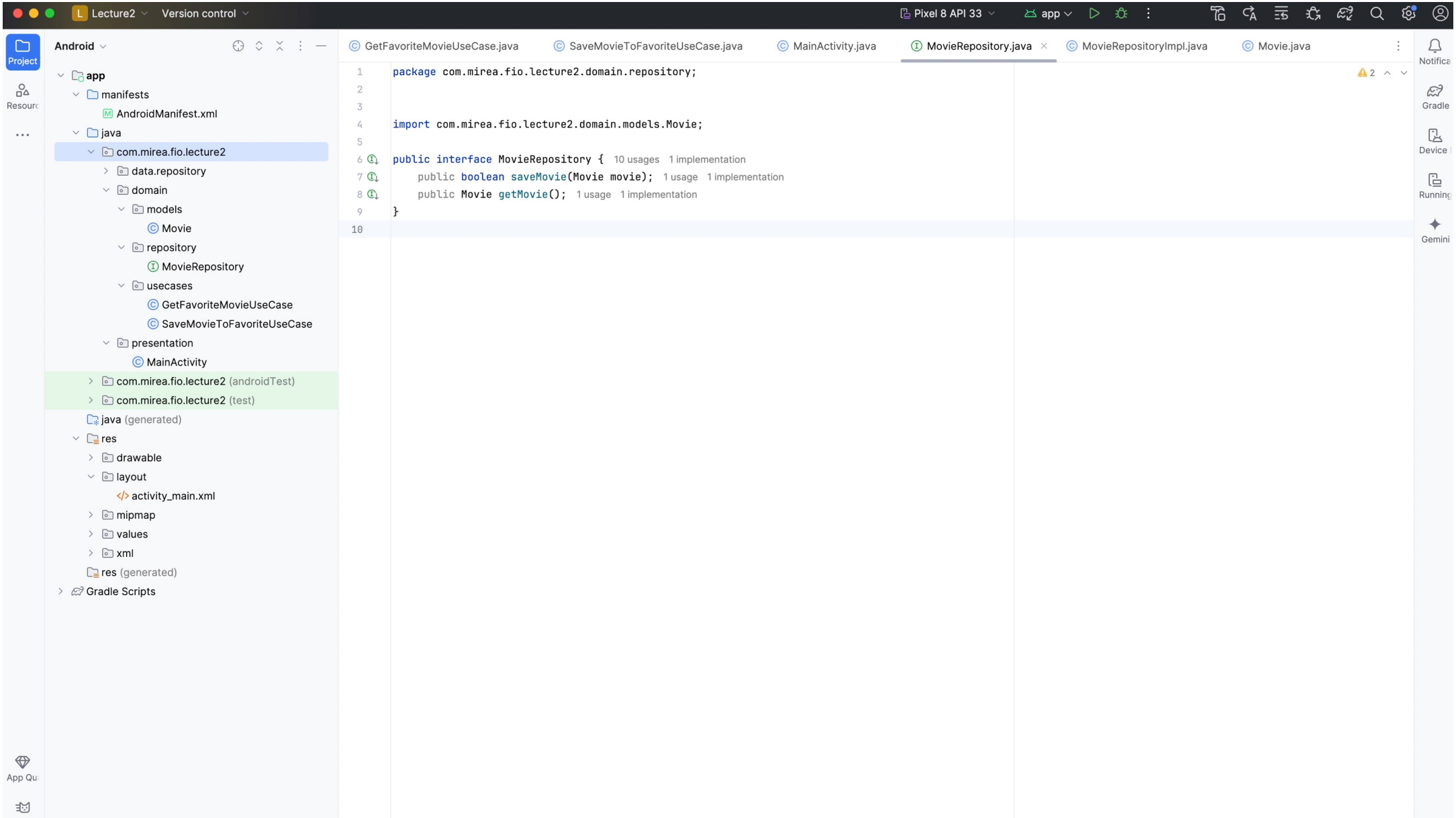
```
public class MovieRepositoryImpl implements MovieRepository {

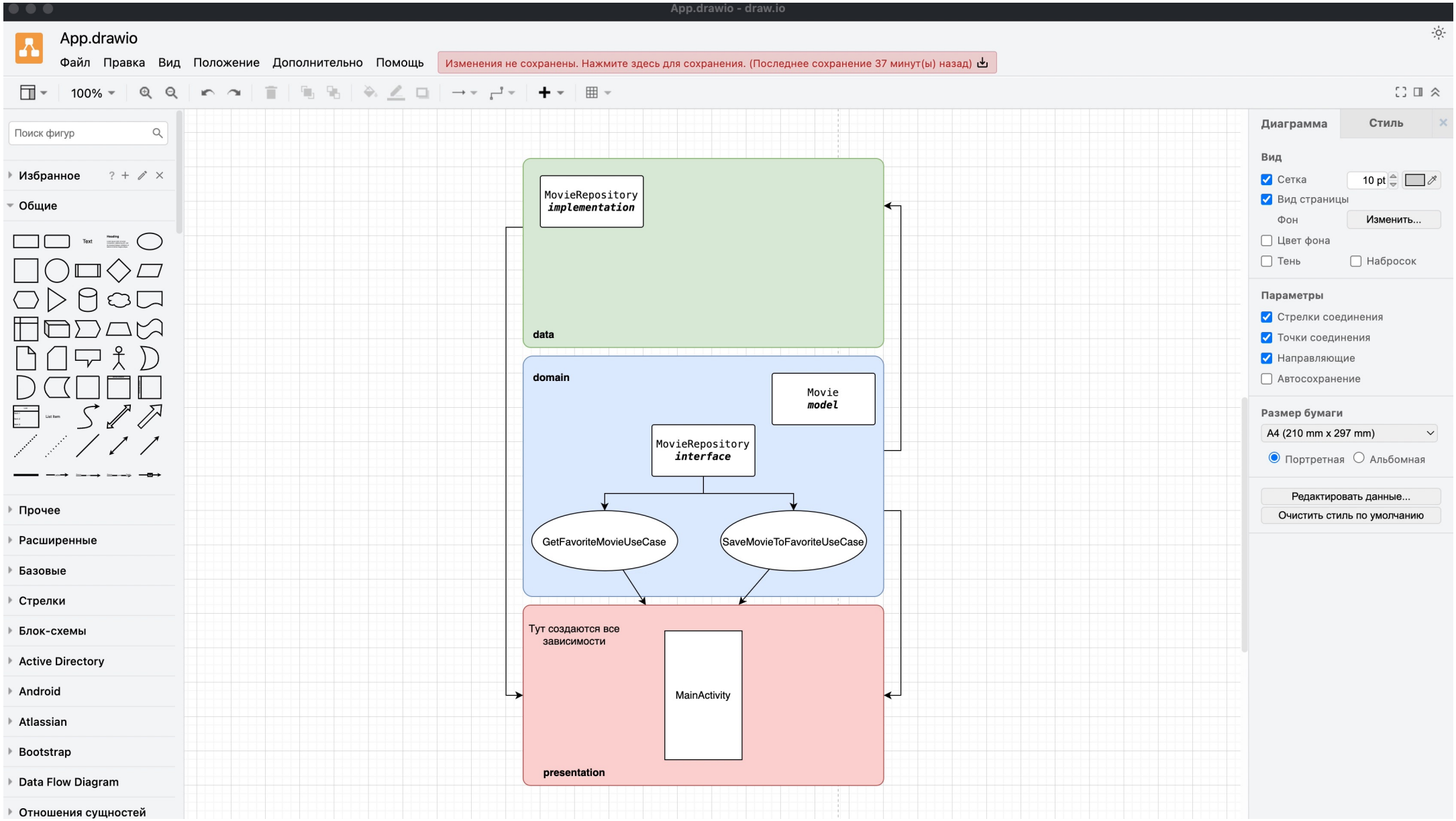
    private static final String SHARED_PREFS_NAME = "shared_prefs_name";
    private static final String KEY = "movie_name";
    private SharedPreferences sharedPreferences;
    private Context context;
    public MovieRepositoryImpl(Context context) {
        this.context = context;
        sharedPreferences = context.getSharedPreferences(SHARED_PREFS_NAME, Context.MODE_PRIVATE);
    }

    @SuppressLint("CommitPrefEdits")
    @Override
    public boolean saveMovie(Movie movie){
        sharedPreferences.edit().putString(KEY, movie.getName()).commit();
        return true;
    }

    @Override
    public Movie getMovie(){
        String movie_name = sharedPreferences.getString(KEY, "unknown");
        return new Movie(1, movie_name);
    }
}
```

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Lecture2
Version control
Pixel 8 API 33
app
Android
GetFavoriteMovieUseCase.java
SaveMovieToFavoriteUseCase.java
MainActivity.java
MovieRepository.java
MovieRepositoryImpl.java
Movie.java
app
manifests
AndroidManifest.xml
java
com.mirea.fio.lecture2
data.repository
domain
models
Movie
repository
MovieRepository
usecases
GetFavoriteMovieUseCase
SaveMovieToFavoriteUseCase
presentation
MainActivity
com.mirea.fio.lecture2 (androidTest)
com.mirea.fio.lecture2 (test)
java (generated)
res
drawable
layout
activity_main.xml
mipmap
values
xml
res (generated)
Gradle Scripts
package com.mirea.fio.lecture2.domain.repository;
import com.mirea.fio.lecture2.domain.models.Movie;
public interface MovieRepository { 10 usages 1 implementation
public boolean saveMovie(Movie movie); 1 usage 1 implementation
public Movie getMovie(); 1 usage 1 implementation
}
Project
Resourc
App Qu
Notifica
Gradle
Device
Running
Gemini


App.drawio - draw.io
App.drawio
Файл Правка Вид Положение Дополнительно Помощь
Изменения не сохранены. Нажмите здесь для сохранения. (Последнее сохранение 37 минут(ы) назад)
100%
Поиск фигур
Избранное
Общие
Прочее
Расширенные
Базовые
Стрелки
Блок-схемы
Active Directory
Android
Atlassian
Bootstrap
Data Flow Diagram
Отношения сущностей
MovieRepository
implementation
data
domain
Movie
model
MovieRepository
interface
GetFavoriteMovieUseCase
SaveMovieToFavoriteUseCase
Тут создаются все зависимости
MainActivity
presentation
Диаграмма
Стиль
Вид
Сетка
10 pt
Вид страницы
Фон
Изменить...
Цвет фона
Тень
Набросок
Параметры
Стрелки соединения
Точки соединения
Направляющие
Автосохранение
Размер бумаги
A4 (210 mm x 297 mm)
Портретная
Альбомная
Редактировать данные...
Очистить стиль по умолчанию

```
package com.mirea.fio.lecture2.data.storage;

import android.content.Context;
import android.content.SharedPreferences;

import com.mirea.fio.lecture2.domain.models.Movie;

public class SharedPrefMovieStorage implements MovieStorage {
    ...

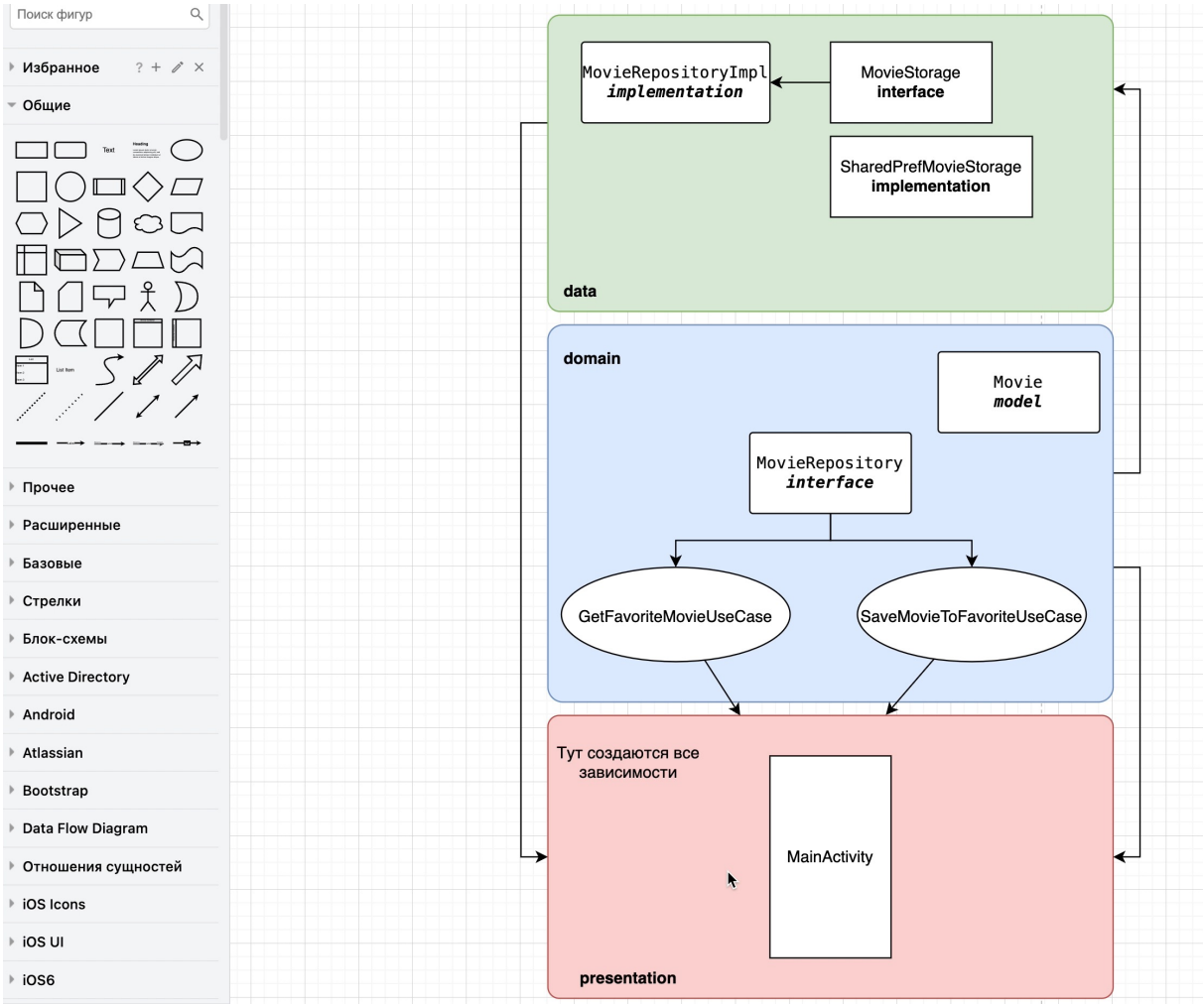
    public SharedPrefMovieStorage(Context context) {
        ...
    }

    @Override
    public Movie get() {
        ...
    }

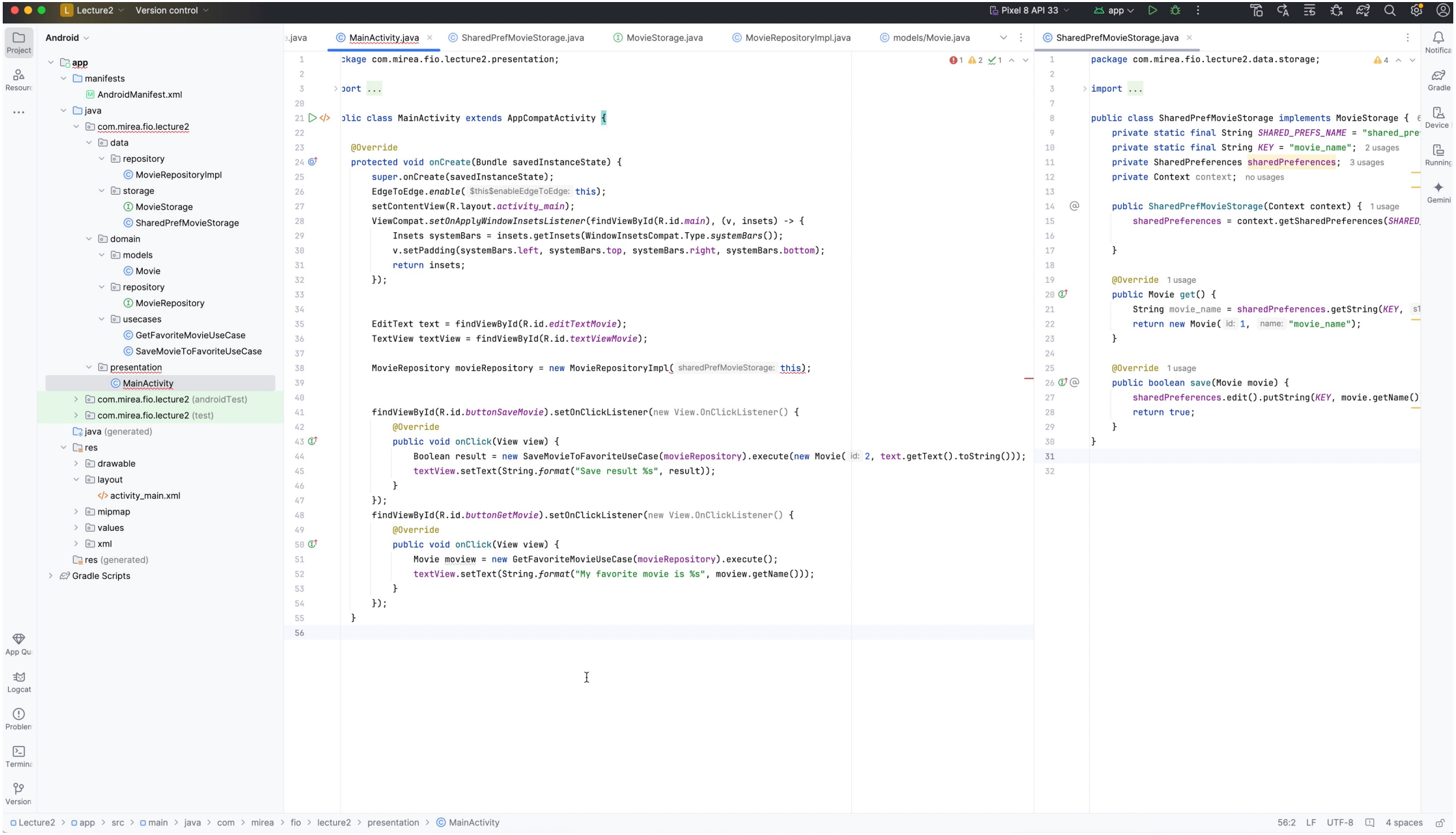
    @Override
    public boolean save(Movie movie) {
        ...
    }
}
```

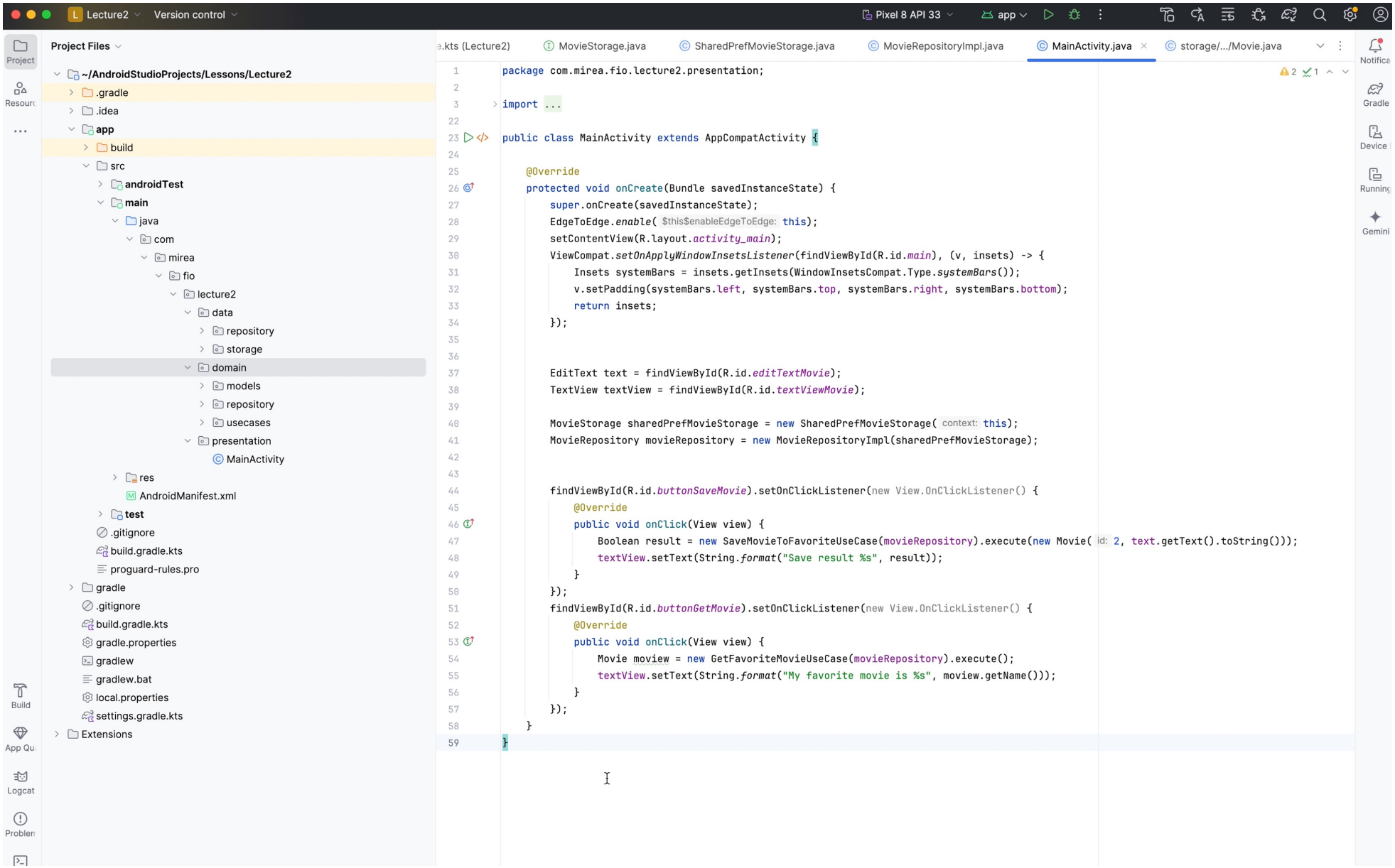
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Чистая архитектура

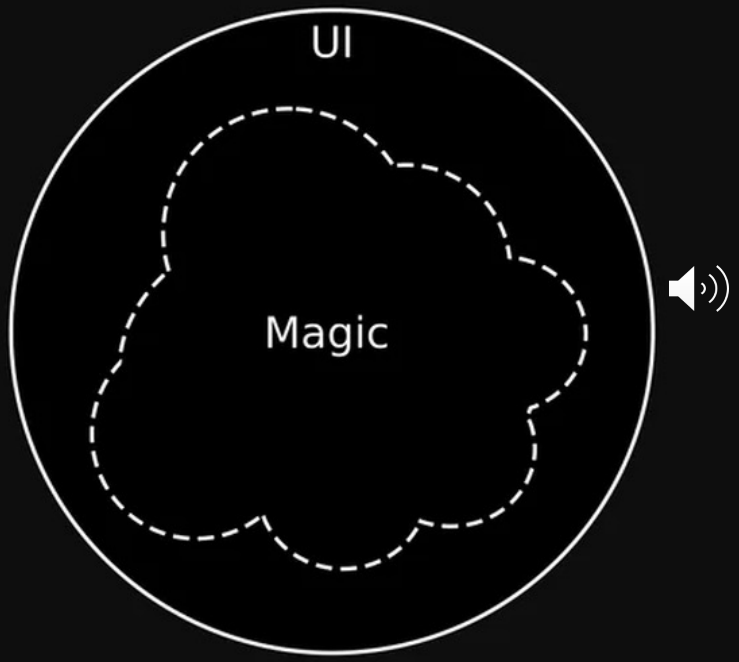


МИРЭА


Lecture2
Version control
Pixel 8 API 33
app
Android
app
manifests
AndroidManifest.xml
java
com.mirea.fio.lecture2
data
repository
MovieRepositoryImpl
storage
MovieStorage
SharedPrefMovieStorage
domain
models
Movie
repository
MovieRepository
usecases
GetFavoriteMovieUseCase
SaveMovieToFavoriteUseCase
presentation
MainActivity
com.mirea.fio.lecture2 (androidTest)
com.mirea.fio.lecture2 (test)
java (generated)
res
drawable
layout
activity_main.xml
mipmap
values
xml
res (generated)
Gradle Scripts
MainActivity.java
SharedPrefMovieStorage.java
MovieStorage.java
MovieRepositoryImpl.java
models/Movie.java
ckage com.mirea.fio.lecture2.presentation;
port ...
blic class MainActivity extends AppCompatActivity {
@Override
protected void onCreate(Bundle savedInstanceState) {
super.onCreate(savedInstanceState);
EdgeToEdge.enable($this$enableEdgeToEdge: this);
setContentView(R.layout.activity_main);
ViewCompat.setOnApplyWindowInsetsListener(findViewById(R.id.main), (v, insets) -> {
Insets systemBars = insets.getInsets(WindowInsetsCompat.Type.systemBars());
v.setPadding(systemBars.left, systemBars.top, systemBars.right, systemBars.bottom);
return insets;
});
EditText text = findViewById(R.id.editTextMovie);
TextView textView = findViewById(R.id.textViewMovie);
MovieRepository movieRepository = new MovieRepositoryImpl(sharedPrefMovieStorage: this);
findViewById(R.id.buttonSaveMovie).setOnClickListener(new View.OnClickListener() {
@Override
public void onClick(View view) {
Boolean result = new SaveMovieToFavoriteUseCase(movieRepository).execute(new Movie(id: 2, text.getText().toString()));
textView.setText(String.format("Save result %s", result));
}
});
findViewById(R.id.buttonGetMovie).setOnClickListener(new View.OnClickListener() {
@Override
public void onClick(View view) {
Movie moview = new GetFavoriteMovieUseCase(movieRepository).execute();
textView.setText(String.format("My favorite movie is %s", moview.getName()));
}
});
}
SharedPrefMovieStorage.java
package com.mirea.fio.lecture2.data.storage;
import ...
public class SharedPrefMovieStorage implements MovieStorage {
private static final String SHARED_PREFS_NAME = "shared_pre
private static final String KEY = "movie_name"; 2 usages
private SharedPreferences sharedPreferences; 3 usages
private Context context; no usages
public SharedPrefMovieStorage(Context context) { 1 usage
sharedPreferences = context.getSharedPreferences(SHARED
@Override 1 usage
public Movie get() {
String movie_name = sharedPreferences.getString(KEY,
return new Movie(id: 1, name: "movie_name");
@Override 1 usage
public boolean save(Movie movie) {
sharedPreferences.edit().putString(KEY, movie.getName()
return true;
Lecture2 › app › src › main › java › com › mirea › fio › lecture2 › presentation › MainActivity
56:2 LF UTF-8 4 spaces


Lecture2
Version control
Pixel 8 API 33
app
Project Files
~/AndroidStudioProjects/Lessons/Lecture2
.gradle
.idea
app
build
src
androidTest
main
java
com
mirea
fio
lecture2
data
repository
storage
domain
models
repository
usecases
presentation
MainActivity
res
AndroidManifest.xml
test
.gitignore
build.gradle.kts
proguard-rules.pro
gradle
.gitignore
build.gradle.kts
gradle.properties
gradlew
gradlew.bat
local.properties
settings.gradle.kts
Extensions
MovieStorage.java
SharedPrefMovieStorage.java
MovieRepositoryImpl.java
MainActivity.java
storage/.../Movie.java
package com.mirea.fio.lecture2.presentation;
import ...
public class MainActivity extends AppCompatActivity {
@Override
protected void onCreate(Bundle savedInstanceState) {
super.onCreate(savedInstanceState);
EdgeToEdge.enable($this$enableEdgeToEdge: this);
setContentView(R.layout.activity_main);
ViewCompat.setOnApplyWindowInsetsListener(findViewById(R.id.main), (v, insets) -> {
Insets systemBars = insets.getInsets(WindowInsetsCompat.Type.systemBars());
v.setPadding(systemBars.left, systemBars.top, systemBars.right, systemBars.bottom);
return insets;
});
EditText text = findViewById(R.id.editTextMovie);
TextView textView = findViewById(R.id.textViewMovie);
MovieStorage sharedPrefMovieStorage = new SharedPrefMovieStorage(context: this);
MovieRepository movieRepository = new MovieRepositoryImpl(sharedPrefMovieStorage);
findViewById(R.id.buttonSaveMovie).setOnClickListener(new View.OnClickListener() {
@Override
public void onClick(View view) {
Boolean result = new SaveMovieToFavoriteUseCase(movieRepository).execute(new Movie(id: 2, text.getText().toString()));
textView.setText(String.format("Save result %s", result));
}
});
findViewById(R.id.buttonGetMovie).setOnClickListener(new View.OnClickListener() {
@Override
public void onClick(View view) {
Movie moview = new GetFavoriteMovieUseCase(movieRepository).execute();
textView.setText(String.format("My favorite movie is %s", moview.getName()));
}
});
}
}
Notifica
Gradle
Device
Running
Gemini
Build
Logcat
Problem

Структурирование кода



Что видят другие



Что видят студенты МИРЭА